

Isokinetic Evaluation of Anterior Cruciate Ligament Reconstruction: Hamstring Versus Patellar Tendon

Thomas R. Carter, M.D., and Samantha Edinger, B.S.

Summary: The purpose of the study was to compare the hamstring and quadriceps isokinetic results 6 months postoperatively in patients having patellar tendon or hamstring anterior cruciate ligament (ACL) reconstruction. The study group was comprised of 106 randomly selected patients who had ACL reconstruction with either autogenous patellar tendon (PT), semitendinosus (ST), or semitendinosus and gracilis (ST/G). Hamstring and quadriceps isokinetic strength were assessed at 180°/sec and 300°/sec with the results of the operatively treated leg expressed as a percent compared with the nonoperative leg. The mean results for knee extension at 180°/sec were 68.3%, 74.3%, and 78.1%; and at 300°/sec were 70.7%, 76.7%, and 81.7% for PT, ST, and ST/G, respectively. The mean results for knee flexion at 180°/sec were 86.1%, 80.6%, and 81.7%; and at 300°/sec were 77.6%, 79.1%, and 75.6% for PT, ST, and ST/G, respectively. No statistically significant differences were found in regard to knee extension or flexion strength when evaluating the different tissue sources. The results show that selection of autogenous hamstring or PT used for ACL reconstruction should not be based solely on the assumption of the tissue source altering the recovery of quadriceps and/or hamstring strength. In addition, a majority of the patients had not achieved adequate strength to safely partake in unlimited activities at 6 months postoperatively. **Key Words:** Anterior cruciate ligament—Reconstruction—Rehabilitation.

The surgical management and rehabilitation programs of anterior cruciate ligament (ACL) injuries have evolved substantially over the last decade. In the past, the procedure was performed in an open fashion, frequently followed by a period of immobilization. Return to sports a year or longer postoperatively was considered the norm. Today, the procedure is routinely done arthroscopically with immediate motion and weight-bearing. It is customary now to allow return to full activities 6 months after the procedure, with some surgeons advocating return to sports as early as 4 months.¹⁻³

Many questions have been raised as a result of these accelerated rehabilitation programs. One area of concern

is that of leg weakness of the involved extremity due to donor-site morbidity. Several investigators have shown significant quadriceps or hamstring weakness that is present beyond the 6-month postoperative time period when using the two common ACL substitutes: autogenous patellar and hamstring tendons.⁴⁻¹⁰ The purpose of this prospective, randomized study was to determine if there was a significant difference in isokinetically measured thigh strength at 6 months in patients having patellar or hamstring grafts, who followed an accelerated rehabilitation program. In addition, analysis as to whether the patients' level of strength was adequate to return to full activities was evaluated.

From Arizona State University, Tempe, Arizona, U.S.A.

Address correspondence and reprint requests to Thomas R. Carter, M.D., Arizona State University, 2700 North Third St, Phoenix, AZ 85004, U.S.A.

© 1999 by the Arthroscopy Association of North America
0749-8063/99/1502-1750\$3.00/0

MATERIALS AND METHODS

A prospective, randomized study was conducted on 120 consecutive patients who underwent arthroscopic-

cally assisted ACL reconstruction. Every third patient sequentially had either the patellar (PT), semitendinosus (ST), or semitendinosus/gracilis (ST/G) tendons used as their tissue for ACL reconstruction. All procedures were performed by the first author (T.R.C.). Of the 120 subjects, 6 were excluded because of their being either Workmens' Compensation cases or having had additional procedures performed other than that of the ACL with or without meniscal surgery. Eight were either lost to follow-up or did not have their leg strength evaluation performed within a 2-week window before or after the 6-month postoperative mark. A total of 106 patients (38 PT, 33 ST, and 35 ST/G) were thus left as the study group.

The three tissue type groups were further subdivided in three categories according to the preinjury level of activities of the patients. Category A was comprised of competitive athletes, including professional, college, or high school level participants. Category B included those who partook in athletics on a regular basis but not on a competitive level. Those patients who only occasionally participated in strenuous activities or sports were included in Category C (Table 1).

The vast majority of the reconstructions were performed for acute ACL tears, i.e., performed within 3 months of the injury. Only 16 were done for chronic insufficiency with the tissue graft distribution being nearly equal, with 6 PT, 6 ST, and 4 ST/G. No significant differences were found using sex, age, leg involved, or meniscal treatment among the three graft groups ($P > .05$).

In an effort to control as many variables as possible, standardized preoperative and rehabilitation programs were followed in all cases. Reconstruction of the knee was delayed until knee swelling was minimal and joint range of motion was near normal. A few days before surgery, the patients were instructed on immediate postoperative treatment, which included such items as patella mobilization techniques, range of motion, and isometric exercises. All reconstructions were performed by one surgeon (T.R.C.) using an arthroscopically aided technique. Interference screws were used to secure the PT grafts with an endoscopic method

used for femoral fixation. Two incisions were used to secure the hamstring tendons with fixation on the femoral side using screw and washer, and on the tibial side with staple fixation in belt buckle fashion.

After surgery, the patients were admitted for overnight observation, with continuous-passive-motion machines used only during their hospital stay. Weight-bearing was allowed to be progressed as tolerated if meniscal repair was not performed, but partial weight bearing was maintained for 4 weeks if meniscal repair was undertaken. Postoperative visits were routinely at 1 week, 4 weeks, and subsequently at monthly intervals until return to full activities, i.e. a minimum of 6 months.

During the first 4 weeks after surgery, range of motion and isometric exercises were continued and regaining their normal gait pattern was stressed to those who were allowed full weight-bearing. At the 4-week mark, the patients were allowed to walk, swim, and bike ride as they felt comfortable. Closed-chain kinetic exercises were initiated as well and gradually progressed. The patients were allowed to run at 16 weeks. If at 26 weeks after surgery they had a stable knee, 80% of their normal knee strength, and acceptable functional testing, then full activities were permitted.

All patients underwent formal physical therapy and followed the same postoperative protocol. However, because of the high incidence of managed care in the Phoenix area, the number of physical therapy visits varied among the patients. On average, they attended physical therapy for 2 to 3 months. As might be expected, the competitive category athletes had the longest duration of supervised rehabilitation with this frequently continuing until return to full activities. However, the quantity of formal physical therapy for each category was similar for the three tissue types.

Once released from supervised therapy, the patients were encouraged to continue their rehabilitation on their own by joining a health club and following a standardized program. Additional counseling pertaining to their rehabilitation was given to the patients by the authors on each follow-up visit.

All patients were tested on a Cybex II isokinetic dynamometer with Data Reduction Computer (Lumex Inc, Ronkonkoma, NY), which was calibrated before the testing period. The testing protocol was standardized to ensure reproducibility and validity.^{11,12} Bilateral seated knee extension/flexion was tested at 6 months (range, 24 to 28 weeks) postoperatively. The patients were instructed on use of the isokinetic machine and allowed to become fully accustomed to

TABLE 1. Patient Distribution by ACL Graft Tissue Source

Activity Category	PT	ST/G	ST
A	17	12	7
B	10	10	8
C	11	13	18

TABLE 2. *Isokinetic Knee Extension Data: Percent Involved/Uninvolved*

Graft Source	Isokinetic Speeds	
	180°/sec	300°/sec
PT	68.3 ± 20.2	70.7 ± 20.9
ST/G	78.1 ± 15.2	81.7 ± 22.1
ST	74.3 ± 20.3	76.7 ± 24.3

the device before testing. They performed a 5-minute warm-up on a stationary bicycle as well. The formal testing consisted of 5 maximal repetitions at 180°/sec and then 15 maximal repetitions at 300°/sec, using the uninvolved extremity first, and then the involved. These speeds were selected for comparison because previous studies have found that strength deficiencies are greater at high speed compared with low speed isokinetic testing.^{12,13} Statistical analysis was determined by analysis of variance test (ANOVA); A probability level of $P < .05$ was considered statistically significant.

RESULTS

Comparison of the isokinetic results among the three graft sources showed general similarities (Tables 2 and 3). Analysis found no statistically significant differences in knee extension or flexion at either speed when looking at the groups as a whole. In addition, when comparing the three tissue groups for each activity category, no significant differences were found in the strength parameters measured.

Table 4 summarizes the number of patients (also expressed as a percentage of total patients) in each activity category and graft-source group who achieved 80% or greater strength in all four of the isokinetic tests when comparing the operative with the nonoperative leg.

DISCUSSION

Strength deficit secondary to graft harvesting has been one area cited as to why neither hamstring nor

TABLE 4. *Number (%) of Patients Having ≥80% Strength in All Parameters by ACL Graft Tissue Source*

Activity Category	PT	ST/G	ST
A	9 (53)	5 (42)	4 (57)
B	6 (60)	5 (50)	2 (25)
C	2 (18)	3 (23)	4 (22)
Total	17 (45)	13 (37)	10 (30)

PTs have been the ideal substitute for the ACL. With the ever increasing efforts to expedite the return not only of the athlete to the playing field, but also the worker back to productive employment, ACL rehabilitation programs have progressed substantially in the recent past. As a result, there has been increased debate as to which of these tendons has less morbidity in regard to strength.¹³⁻¹⁷

The ability to limit this loss, as well as to regain strength deficiencies, has definitely been shown to be related to the aggressiveness of postoperative treatment. Natri et al.⁶ found continued quadriceps strength deficits of 20% in patients with PT graft when evaluated at an average of 4.3 years after surgery. The postoperative program was conservative by today's standards, including 6 to 8 weeks of brace and crutch use with return to sports as long as 1 year postoperatively.

Following similar postoperative programs, other authors have also reported appreciable strength deficiencies. Sachs et al.⁸ found the mean quadriceps strength to be only 60.8% with PT grafts at 1 year postoperatively. Marder et al.⁵ found the operatively treated leg to have on average 17% deficiency in 20 patients followed-up for a minimum of 2 years who underwent ACL reconstruction with ST/G grafts. Fortunately most authors report resolution of these strength deficiencies with time.¹⁷⁻¹⁹

Studies in which more aggressive rehabilitation was permitted have demonstrated improved leg strength with both graft types. Interesting though, in many patients there was still noticeable lack of strength in the involved extremity at the 4 to 6 months after surgery. Wilk et al.¹⁰ found that patients undergoing ACL reconstruction with PT had an average of 20% to 25% deficit in total work when measured at 180°/sec and 300°/sec at 6 months postoperatively. However, it should be noted that all patients were considered recreational athletes. Shelbourn and Nitz³ followed their accelerated rehabilitation program and reported an average of 25% quadriceps peak torque deficiency at 180°/sec between 4 and 6 months postoperatively.

TABLE 3. *Isokinetic Knee Flexion Data: Percent Involved/Uninvolved*

	Isokinetic Speeds	
	180°/sec	300°/sec
PT	86.1 ± 21.9	77.6 ± 25.9
ST/G	81.7 ± 18.5	75.6 ± 19.0
ST	80.6 ± 28.3	79.1 ± 27.4

Although Howell and Taylor¹ did not report isokinetic testing in their study on allowing hamstring reconstruction patients to return to full activities at 4 months, they did find continued improvement of functional testing beyond the 4-month mark.

The results of our study concur with the isokinetic studies previously reported in the literature regarding the quadriceps strength for PTs following an accelerated rehabilitation program. Unfortunately, comparison pertaining to knee extension strength for those having hamstring reconstruction and following an accelerated rehabilitation program could not be made due to there being a void pertaining specifically to isokinetic testing at the 6-month mark. Regardless, both graft sources showed appreciable deficiencies in leg extension with neither having a clear-cut benefit in regard to strength measurements. In similar fashion, while one may intuitively expect knee flexion strength to have a greater deficiency in hamstring donors, no significant differences were found to justify the selection of one tissue source over the other based on this parameter.

The isokinetic results also showed that an appreciable quantity of patients had not regained adequate strength to be released to full activities without concern. As has been shown by Daniel et al.²⁰ and followed by others, the ACL-reconstructed leg should have 80% or greater return to the strength of the normal limb as one criterion before release to full activities.^{3,4,20} We were particularly intrigued to find only half of the competitive athletes had achieved this goal by 6 months after surgery.

CONCLUSIONS

The strength deficiencies found in our study concur with what has previously been shown in regard to ACL reconstruction and harvesting of autogenous tissue for the graft. No evidence was found in regard to leg strength as a basis for selecting either PT, ST, or ST/G tendons as the optimal graft. In addition, regardless of which of these tissues is used, a considerable percentage of patients continue to have strength deficits at 6 months. Therefore, the authors strongly encourage surgeons to educate their patients who are to have ACL reconstructions that, theoretically, they may return to sports 4 to 6 months postoperatively, but that frequently leg strength is not adequate at that time to do so without running the risk of reinjuring the knee.

REFERENCES

1. Howell SM, Taylor MA. Brace-free rehabilitation, with early return to activity, for knees reconstructed with a double-looped semitendinosus and gracilis graft. *J Bone Joint Surg Am* 1996;78:814-825.
2. MacDonald PB, Hedder D, Pacin O, et al. Effects of an accelerated rehabilitation program after anterior cruciate ligament reconstruction with combined semitendinosus-gracilis autograft and ligament augmentation device. *Am J Sports Med* 1995;23:588-592.
3. Shelbourn KD, Nitz P. Accelerated rehabilitation after anterior cruciate ligament reconstruction. *Am J Sports Med* 1990;18:292-299.
4. Hugel M, Indelicato P. Trends in rehabilitation following anterior cruciate ligament reconstruction. *Clin Sports Med* 1988;7:801-811.
5. Marder RA, Raskind JR, Carrol M. Prospective evaluation of arthroscopically assisted anterior cruciate ligament reconstruction. *Am J Sports Med* 1991;19:478-483.
6. Natri A, Jarvinen M, Latuala K, et al. Isokinetic muscle performance after anterior cruciate ligament surgery. *Int J Sports Med* 1996;17:223-228.
7. Rosenberg TD, Franklin JL, Baldwin, GN, et al. Extensor mechanism function after patellar tendon graft harvest for anterior cruciate ligament reconstruction. *Am J Sports Med* 1992;20:519-526.
8. Sachs RA, Daniel DM, Stone ML, et al. Patellofemoral problems after anterior cruciate ligament reconstruction. *Am J Sports Med* 1989;17:760-765.
9. Yasuda K, Tsujino J, Ohkoshi Y, Tanabe Y, et al. Graft site morbidity with autogenous semitendinosus and gracilis tendons. *Am J Sports Med* 1995;23:706-714.
10. Wilk KE, Keirns MA, Andrews JR, et al. Anterior cruciate ligament reconstruction rehabilitation: A six-month followup of isokinetic testing in recreational athletes. *Isokinet Exerc Sci* 1991;1:36-43.
11. Wilk KE. Dynamic muscle testing. In: Amundsen LR, ed. *Muscle strength testing, instrumented and non-instrumented systems*. New York: Churchill Livingstone, 1990.
12. Gieck JH, Saliba EN. The athletic trainer and rehabilitation. In: Kulund DN, ed. *The injured athlete*. Ed 2. Philadelphia: JB Lippincott, 1988;165-234.
13. Kannus P, Latuala K, Jarvinen M. Thigh muscle strengths in the anterior cruciate ligament deficient knee: Isokinetic and isometric long-term results. *J Orthop Sports Phys Ther* 1987;9:223-227.
14. Franke K. Clinical experience in 130 cruciate ligament reconstructions. *Orthop Clin North Am* 1976;7:191-193.
15. Noyes FR, Butler DL, Grood ES, et al. Biomechanical analysis of human ligament grafts used in knee-ligament repairs and reconstructions. *J Bone Joint Surg Am* 1984;66:344-352.
16. Paulos L, Noyes FR, Grood E, et al. Knee rehabilitation after anterior cruciate ligament reconstruction and repair. *Am J Sports Med* 1981;9:140-149.
17. Bach BR Jr, Jones GT, Sweet FA, et al. Arthroscopy-assisted anterior cruciate ligament reconstruction using patellar tendon substitution. *Am J Sports Med* 1994;22:758-767.
18. Lephart SM, Kocher MS, Harner CD, et al. Quadriceps strength and functional capacity after anterior cruciate ligament reconstruction: Patellar tendon autograft versus allograft. *Am J Sports Med* 1993;21:738-743.
19. Sadedmi SR, Frogameni AD, Fenton PJ. Comparison of perioperative morbidity of anterior cruciate ligament autografts versus allografts. *Arthroscopy* 1993;9:519-524.
20. Daniel D, Malcolm L, Stone ML, et al. Quantification of knee stability and function. *Contemp Orthop* 1982;5:83-92.